

Anderson localisation with self-avoiding-walks

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1 Introduction

The Anderson localization model was introduced by the physicist P.W. Anderson in 1958 to explain the quantum mechanical effects of disorder in materials. Anderson discusses the behaviour of electrons in dirty crystals, which is the quantum mechanical analogue of a random walk in a random environment. The model considers a d -dimensional grid, in which each site represents an atom. From an heuristical point of view, we also consider electrons that jump from atom to atom and are subject to an external random potential, which influences their motion in the material. When these particles cannot jump, it indicates the absence of quantum transport. This phenomenon is called Anderson localisation. This model deals with conditions for which the electrons stay trapped in a specific region and don't generate electric conduction, defining an insulator. This is in sharp contrast to the behaviour in ideal crystals, which are always conductors.

Anderson's work, and that of N.F.Mott and J.H. van Vleck, won the 1977 physics Nobel prize for "their fundamental theoretical investigations of the electronic structure of magnetic and disordered systems". Since then, the study of random operators has become an important field of mathematical physics, which has led to a tremendous amount of research activity and many mathematical results.

The two important methods to prove Anderson localization are the *multiscale analysis method* (MSA), developed by Froehlich and Spencer [1] in 1983, and the *fractional moment method* (FMM) by Aizenman and Molchanov in 1993. We will deal with the second one, which is more elementary and gives stronger results on *dynamical localization* [Chapter 2.2].

structure of the thesis

Prerequisites

It is useful to have some familiarity with measure theory, basic probabilistic

concepts such as independence and continuous distributions, and foundations of the theory of linear operators in Hilbert spaces, up to the spectral theorem for self adjoint operators and consequences (as spectral types).

1.1 The Anderson Model

Transport phenomena in a disordered material are often described via discrete random Schrödinger operator. In quantum mechanics, it corresponds to the classical total energy $E = T + V$, decomposing into kinetic energy T and potential energy V . On the lattice $\mathbb{Z}^d$, it takes the form of an infinite random matrix called the *Hamiltonian*:

$$H_\omega = -\Delta + \lambda V_\omega.$$

where $-\Delta$ is a deterministic matrix and V_ω is a diagonal matrix with random entries. This operator acts on the Hilbert space

$$\ell^2(\mathbb{Z}^d) = \{u : \mathbb{Z}^d \rightarrow \mathbb{C} : \sum_{n \in \mathbb{Z}^d} |u(n)|^2 < \infty\}$$

with inner product $\langle u, v \rangle = \sum_n \overline{u(n)}v(n)$.

1.1.1 The operator

The Discrete Laplacian is the discrete analogue of the usual negative Laplacian $(-\sum_j \partial^2 / \partial x_j^2)$. For the functions $u \in \ell^2(\mathbb{Z}^d)$ it is defined as:

$$-\Delta u(x) := \sum_{k \in \mathbb{Z}^d, |x-k|=1} u(x) - u(k) \quad \text{for } x \in \mathbb{Z}^d.$$

where $|\cdot|$ denotes the ℓ^2 norm.

The discrete Laplacian represents the kinetic term and connects neighbour sites on the lattice, such that the electron can transition from one to another.

For each site x , there are $2d$ possible k such that $|x - k| = 1$. Thus, we

can define an infinite matrix such that:

$$(-\Delta)_{x,k} = \begin{cases} 2d & \text{if } x = k \quad (\text{on the diagonal}) \\ -1 & \text{if } |x - k| = 1 \\ 0 & \text{otherwise} \end{cases}$$

For $u \in \ell^2(\mathbb{Z}^d)$, $-\Delta u$ is summable in ℓ^2 , bounded, and self-adjoint.

The first claim (summability) is shown by:

$$\begin{aligned} \sum_{x \in \mathbb{Z}^d} |\Delta u(x)|^2 &= \sum_{x \in \mathbb{Z}^d} \left| \sum_{k \in \mathbb{Z}^d, |x-k|=1} (u(x) - u(k)) \right|^2 \\ &\leq 2d \sum_{x \in \mathbb{Z}^d} \left(\sum_{|x-k|=1} |u(x) - u(k)|^2 \right) \\ &\leq 2d \sum_{x \in \mathbb{Z}^d} \left(2 \sum_{|x-k|=1} (|u(x)|^2 + |u(k)|^2) \right) \\ &\leq 2d \cdot 2d \cdot 4 \|u\|_2^2 < \infty \end{aligned}$$

where we used the bound over the number of neighbors, triangle inequality, Cauchy Schwarz and the invariance under translation of the ℓ^2 norm. The boundedness comes from:

$$\frac{\|\Delta u\|_2^2}{\|u\|_2^2} \leq 16d^2 \implies \|\Delta\|_2 \leq 4d.$$

To prove self-adjointness, we only need to show symmetry (given $-\Delta$ is bounded). For $u, v \in \ell^2(\mathbb{Z}^d)$ and $k \in \mathbb{Z}^d$:

$$\begin{aligned} \langle u, -\Delta v \rangle &= \sum_{x \in \mathbb{Z}^d} \overline{u(x)} (-\Delta v(x)) = \sum_{x \in \mathbb{Z}^d} \overline{u(x)} \left(\sum_{|x-k|=1} v(x) - v(k) \right) \\ &= \sum_{x \in \mathbb{Z}^d} \left(\overline{u(x)} 2dv(x) - \sum_{|x-k|=1} \overline{u(x)} v(k) \right) = \sum_{x \in \mathbb{Z}^d} 2d \overline{u(x)} v(x) - \sum_{x \in \mathbb{Z}^d} \sum_{|x-k|=1} \overline{u(k)} v(x) \\ &= \sum_{x \in \mathbb{Z}^d} v(x) \left(\sum_{|x-k|=1} \overline{u(x)} - \overline{u(k)} \right) = \sum_{x \in \mathbb{Z}^d} v(x) \overline{(-\Delta u(x))} = \langle -\Delta u, v \rangle. \end{aligned}$$

The spectrum of the bounded, self-adjoint operator $-\Delta$ is the closed, real-valued interval $[0, 4d]$. In this range, we find the levels of kinetic energy that an electron owns. In order to understand this result better, let's consider the discrete Fourier transform:

$$\mathcal{F}: \ell^2(\mathbb{Z}^d) \longrightarrow L^2([0, 2\pi)^d)$$

such that:

$$(\mathcal{F}u)(x) := \lim_{N \rightarrow \infty} (2\pi)^{-\frac{d}{2}} \sum_{|n| \leq N} u(n) e^{-ix \cdot n}, \quad x \in [0, 2\pi)^d.$$

For simplicity of the calculations we don't consider the limit and just sum over all $n \in \mathbb{Z}^d$ and u summable.

$$\begin{aligned} (\mathcal{F}(-\Delta u))(x) &= (2\pi)^{-\frac{d}{2}} \sum_{n \in \mathbb{Z}^d} (-\Delta u)(n) e^{-ix \cdot n} \\ &= (2\pi)^{-\frac{d}{2}} \sum_{n \in \mathbb{Z}^d} \left(\sum_{|k-n|=1} (u(n) - u(k)) \right) e^{-ix \cdot n} \\ &= (2\pi)^{-\frac{d}{2}} \sum_{n \in \mathbb{Z}^d} \left(2d u(n) - \sum_{|k-n|=1} u(k) \right) e^{-ix \cdot n} \\ &= 2d (\mathcal{F}u)(x) - (2\pi)^{-\frac{d}{2}} \sum_{n \in \mathbb{Z}^d} \sum_{|k-n|=1} u(k) e^{-ix \cdot n} \\ &= 2d (\mathcal{F}u)(x) - (2\pi)^{-\frac{d}{2}} \sum_{k \in \mathbb{Z}^d} u(k) \sum_{|m|=1} e^{-ix \cdot (k+m)} \quad (\text{for } m = n - k) \\ &= 2d (\mathcal{F}u)(x) - \left((2\pi)^{-\frac{d}{2}} \sum_{k \in \mathbb{Z}^d} u(k) e^{-ix \cdot k} \right) \sum_{|m|=1} e^{-ix \cdot m} \\ &= 2d (\mathcal{F}u)(x) - (\mathcal{F}u)(x) \sum_{j=1}^d (e^{-ix_j} + e^{ix_j}) \\ &= \left(2 \sum_{j=1}^d (1 - \cos(x_j)) \right) (\mathcal{F}u)(x). \end{aligned}$$

We obtain the multiplication operator:

$$\mathcal{F}\Delta\mathcal{F}^{-1} = 2 \sum_{j=1}^d (1 - \cos(x_j))$$

on $L^2([0, 2\pi)^d)$ in the variable $x = (x_1, \dots, x_d)$. We notice that the Laplacian doesn't have eigenvalues in $\ell^2(\mathbb{Z}^d)$. Another similarity of Δ with the continuum Laplacian is that it has plane waves as generalized eigenfunctions. To see this, let $x \in [0, 2\pi)^d$ and set

$$\phi_x(n) := e^{-in \cdot x}$$

While $\phi_x \notin \ell^2(\mathbb{Z}^d)$, Δ acts on it as

$$\begin{aligned} (\Delta\phi_x)(n) &= 2d\phi_x(n) - \left(\sum_{j=1}^d 2\cos(x_j) \right) \phi_x(n) \\ &= \left(2 \sum_{j=1}^d (1 - \cos(x_j)) \right) \phi_x. \end{aligned} \quad \text{program@epstopdf}$$

Thus ϕ_x is a bounded generalized eigenfunction of Δ to the spectral value $2 \sum_{j=1}^d (1 - \cos(x_j))$.

The Random Potential.

Definition 1.1 (i.i.d. random variables). $(\omega_n)_{n \in \mathbb{Z}^d}$ are identically distributed if there exists a Borel probability measure μ on $\mathbb{R}$ such that $\mathbb{P}(\omega_n \in A) = \mu(A)$ for all $n \in \mathbb{Z}^d$ and Borel sets $A \subset \mathbb{R}$. $(\omega_n)_{n \in \mathbb{Z}^d}$ are independent if

$$\mathbb{P}(\omega_{n_1} \in A_1, \dots, \omega_{n_l} \in A_l) = \prod_{k=1}^l \mathbb{P}(\omega_{n_k} \in A_k) = \prod_{k=1}^l \mu(A_k)$$

for each finite subset $\{n_1, \dots, n_l\}$ of $\mathbb{Z}^d$ and arbitrary Borel sets $A_1, \dots, A_l \subset \mathbb{R}$.

We assume that the distribution μ of ω_v is absolutely continuous with

density ρ where ρ is bounded with compact support, i.e.,

$$\mu(B) = \int_B \rho(u) du \quad \text{for } B \subset \mathbb{R} \text{ Borel, } \rho \in L_0^\infty(\mathbb{R}).$$

Physically, V_ω represents the random electric potential created by nuclei at the sites $n \in \mathbb{Z}^d$ **ovviamente il testo da modificare in base al resto delle notazioni**.

$\lambda > 0$ is the disorder parameter

- Definition
- heuristic role generic explanation
- explanation for 0 and very big to give an idea.

1.1.2 The spectrum.

- in $\mathbb{R}$
- bounded $[0, 4d] + \text{supp}\mu$
- spectral division (any use of the spectral theorem should be sent to the Appendix)
- RAGE theorem (or maybe directly in the Appendix or chapter 2)

Dalla teoria spettrale per operatori autoaggiunti su uno spazio di Hilbert $\mathcal{H}$, lo spazio si decompone univocamente nella somma diretta di sottospazi invarianti ortogonali:

$$\mathcal{H} = \mathcal{H}_{pp} \oplus \mathcal{H}_{sc} \oplus \mathcal{H}_{ac} \tag{1.1}$$

a cui corrispondono rispettivamente lo spettro puramente puntuale $\sigma_{pp}(H)$, singolarmente continuo $\sigma_{sc}(H)$ e assolutamente continuo $\sigma_{ac}(H)$, con $\sigma(H) = \sigma_{pp}(H) \cup \sigma_{sc}(H) \cup \sigma_{ac}(H)$.

L'interpretazione fisica di questi sottospazi è chiarita dal fondamentale **Teorema RAGE** (Ruelle-Amrein-Georgescu-Enss):

Theorem 1.2 (RAGE). *Suppose H is a Schrödinger operator acting on $\ell^2(\mathbb{Z}^d)$. Then:*

- (i) $\phi \in \mathcal{H}_{pp}$ if and only if

$$\lim_{R \rightarrow \infty} \sup_{t \in \mathbb{R}} \|\chi_{(|x| > R)} e^{-itH} \phi\| = 0.$$

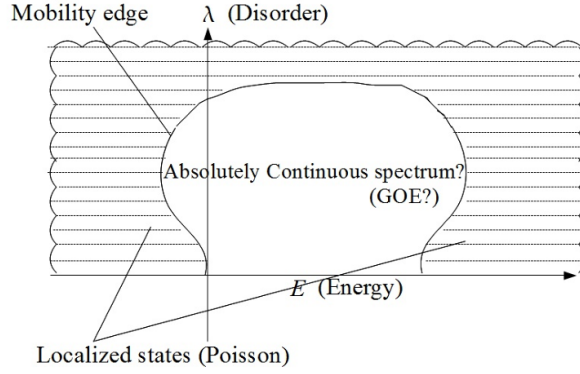


Figure 1: Aizenman figure

(ii) $\psi \in \mathcal{H}_c = \mathcal{H}_{ac} \oplus \mathcal{H}_{sc}$ if and only if

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \|\chi_{(|x| \leq R)} e^{-itH} \psi\| dt = 0$$

for all $R > 0$.

Here $\chi_{(|x| \leq R)}$ and $\chi_{(|x| > R)}$ denote the characteristic functions of the ball $\{|x| \leq R\}$ and the complement of the ball, respectively.

1.2 Main results

In this section one could write first of all the main theorem and the spectral results. starting maybe with:

The aim of this thesis is to prove localisation on the whole spectrum of H_ω
...

Poi scrivo il teorema finale con spiegazione che la dimostrazione avverrà nel capitolo 5.

potrebbe essere un'idea anche scrivere una piccola parentesi sulla differenza dei risultati con Frohlich e spencer e con Aizenman e Molchanov.

One of the main results concerns the values of λ

- $\lambda \gg 1$ and $\lambda = 0$
- general case for $d = 1, d \geq 2$

2 Localisation properties

We understood what localisation means from an heuristical point of view. Mathematically, it can be described from spectral and dynamical properties.

2.1 spectral localisation

Definition 2.1 (Spectral Localization). random Schrödinger operator $H = H_\omega$ exhibits **spectral localization** if H_ω almost surely has only pure point spectrum, that is,

$$\sigma(H_\omega) = \sigma_{pp}(H_\omega) \quad \text{and} \quad \sigma_c(H_\omega) = \emptyset \quad \text{almost surely.}$$

Furthermore the eigenfunctions corresponding to all eigenvalues decay exponentially: for almost all ω , H_ω has a complete set of eigenfunctions $(\phi_{\omega,n})_{n \in \mathbb{N}}$ satisfying

$$|\phi_{\omega,n}(x)| \leq C_{\omega,n} e^{-\mu|x-x_{n,\omega}|}$$

where $\mu > 0$ is the localization length parameter, $x_{n,\omega}$ are the centers of localization, and $C_{\omega,n} < \infty$ is a finite constant.

2.1.1 Absence of transport

$$\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t) = \left\| |x| e^{-itH_\omega} P_{H_\omega \in [a,b]} \phi_0 \right\|^2 \quad (5)$$

If the electrons with energies in $[a, b]$ move ballistically with average velocity v_{av} , then roughly $x(t) \sim v_{av}t$ for large times. Hence $\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t)$ will be proportional to t^2 for large t .

On the other hand, if the electrons in the energy interval $[a, b]$ are localized, then it should be natural to expect that $\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t)$ is bounded uniformly in t . It is known (Simon, 1990) that pure point spectrum implies absence of ballistic motion,

$$\lim_{t \rightarrow \infty} \frac{\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t)}{t^2} = 0$$

for all compactly supported initial conditions ϕ_0 , as soon as the spectrum in $[a, b]$ is pure point. However, del Rio et al. (1996) constructed examples of (non-random) one-dimensional Schrödinger operators with pure point spec-

trum for all energies and exponentially localized eigenfunctions for which

$$\limsup_{t \rightarrow \pm\infty} \frac{\langle x^2 \rangle_{\phi_0}(t)}{t^{2-\delta}} = \infty \quad \text{for all } \delta > 0, \quad (6)$$

for a large class of compactly supported initial conditions ϕ_0 . *Il fallimento risiede nella libertà delle costanti $C_{\omega,n}$: se $C_{\omega,n}$ cresce incontrollatamente con n , le code delle autofunzioni possono sovrapporsi su scale spaziali arbitrariamente grandi.*

2.2 dynamical localisation

Definition 2.2 (Dynamical localisation).

$$\sup_{t \in \mathbb{R}} \| |X|^p e^{-itH_\omega} \psi \| < \infty \quad \text{almost surely}, \quad (2.1)$$

where the position operator $|X|$ is defined by $(|X|\phi)(n) = |n|\phi(n)$ and for all φ with compact support..

Definition 2.3 (Exponential localisation).

$$\mathbb{E} \left(\sup_{t \in \mathbb{R}} |\langle e_j, e^{-itH_\omega} e_k \rangle| \right) \leq C e^{-\mu|j-k|} \quad (2.2)$$

Remark 2.4. Dynamical localization in the form (2.2) is a strong form of asserting that solutions of the time-dependent Schrödinger equation $H_\omega \phi(t) = i\partial_t \phi(t)$ are staying localized in space, uniformly for all times, and thus shows the absence of quantum transport.

Proposition: *Exponential localization implies that all moments of the position operator are bounded in time, i.e., dynamical localisation*

Proof. To see how (2.1) follows from (2.2), we assume that $\psi(k) = 0$ for $|k| > R$. Then

$$\begin{aligned} \| |X|^p e^{-itH_\omega} \psi \|^2 &= \sum_j |\langle e_j, |X|^p e^{-itH_\omega} \psi \rangle|^2 \\ &= \sum_j |j|^{2p} \left| \sum_{|k| \leq R} \langle e_j, e^{-itH_\omega} e_k \rangle \psi(k) \right|^2 \\ &\leq \sum_j \sum_{|k| \leq R} |j|^{2p} |\langle e_j, e^{-itH_\omega} e_k \rangle|^2 \|\psi\|^2, \end{aligned}$$

where the last step used the Cauchy-Schwarz inequality. We can drop the square from $|\langle e_j, e^{-ith_\omega} e_k \rangle|^2$ (as this number is bounded by 1) and then take expectations to get

$$\begin{aligned} \mathbb{E} \left(\sup_t \| |X|^p e^{-ith_\omega} \psi \|^2 \right) &\leq \sum_j \sum_{|k| \leq R} |j|^{2p} \mathbb{E} \left(\sup_t |\langle e_j, e^{-ith_\omega} e_k \rangle| \right) \|\psi\|^2 \\ &\leq C \sum_j \sum_{|k| \leq R} |j|^{2p} e^{-\mu|j-k|} \|\psi\|^2 \\ &< \infty. \end{aligned}$$

□

2.3 From dynamical to spectral localisation

Proposition 2.5. *Suppose that dynamical localization in the form*

$$\mathbb{E} \left(\sup_{t \in \mathbb{R}} |\langle e_j, e^{-ith_\omega} \chi_I(h_\omega) e_k \rangle| \right) \leq C e^{-\mu|j-k|} \quad (2.3)$$

holds in an open interval I . Then h_ω almost surely has pure point spectrum in I .

Proof. For a discrete Schrödinger operator $h = h_0 + V$ in $\ell^2(\mathbb{Z}^d)$ let $P_{\text{cont}}(h)$ be the projection onto its continuous spectral subspace. Then the RAGE-Theorem says that for every $\psi \in \ell^2(\mathbb{Z}^d)$,

$$\|P_{\text{cont}}(h) \chi_I(h) \psi\|^2 = \lim_{R \rightarrow \infty} \lim_{T \rightarrow \infty} \int_0^T \frac{dt}{T} \|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h) \psi\|^2. \quad (2.4)$$

If ψ has finite support, say $\text{supp } \psi \subset \{|x| \leq r\}$, then

$$\begin{aligned} \|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h) \psi\|^2 &\leq \|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h) \chi_{\{|x| \leq r\}}\|^2 \|\psi\|^2 \\ &\leq \|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h) \chi_{\{|x| \leq r\}}\| \|\psi\|^2 \\ &\leq \sum_{|x| \geq R, |y| \leq r} |\langle e_x, e^{-ith} \chi_I(h) e_y \rangle| \|\psi\|^2, \end{aligned}$$

where dropping a square is allowed as $\|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h) \chi_{\{|x| \leq r\}}\| \leq 1$.

Taking expectations in (2) implies, after using Fatou and Fubini,

$$\mathbb{E}(\|P_{\text{cont}}(h_\omega)\chi_I(h_\omega)\psi\|^2) \leq \lim_{R \rightarrow \infty, T \rightarrow \infty} \int_0^T \frac{dt}{T} \sum_{|x| \geq R, |y| \leq r} \mathbb{E}(|\langle e_x, e^{-ith_\omega} \chi_I(h_\omega) e_y \rangle|) \|\psi\|^2. \quad (2.5)$$

By (2.4) we have $\mathbb{E}(|\langle e_x, e^{-ith_\omega} \chi_I(h_\omega) e_y \rangle|) \leq C e^{-\mu|x-y|}$ uniformly in t , which bounds the right hand side of (2.6) by

$$\leq \lim_{R \rightarrow \infty} \tilde{C} \sum_{|x| \geq R, |y| \leq r} e^{-\mu|x-y|} = 0.$$

We conclude that $P_{\text{cont}}(h_\omega)\chi_I(h_\omega)\psi = 0$ for almost every ω and every ψ of finite support. The latter are dense in $\ell^2(\mathbb{Z}^d)$ and thus $P_{\text{cont}}(h_\omega)\chi_I(h_\omega) = 0$ almost surely, meaning that the spectrum in I is pure point. $\square$

3 Fractional moments

In this section we are going to see the important lemmas and theorems that help us proving localisation. In particular we see the power of the fractional moment method.

3.1 The green function

The *resolvent operator* is defined as:

$$G(z) = (H - z)^{-1}$$

3.2 Localisation at large disorder

Theorem 3.1 (Aizenman–Molchanov, 1993). *Let $0 < s < 1$. Then there exists $\lambda_0 > 0$ such that for $\lambda \geq \lambda_0$ there are $C < \infty$ and $\mu > 0$ with*

$$\mathbb{E}(|G_{\omega,\lambda}(x, y; z)|^s) \leq Ce^{-\mu|x-y|} \quad (3.1)$$

uniformly in $x, y \in \mathbb{Z}^d$ and $z \in \mathbb{C} \setminus \mathbb{R}$. In particular, if λ_{And} satisfies $\lambda_{\text{And}} = \mu_d e \ln \lambda_{\text{And}}$ then for all $\lambda > \lambda_{\text{And}}, \lambda > e$ and $\epsilon > 0$ we have

$$\mathbb{E}\left(|G^{(\Lambda)}(x, y; z)|^{1-\frac{1}{\ln \lambda}}\right) \leq \beta(\lambda) K_\epsilon \ln \lambda e^{-m_\epsilon(\lambda)|x-y|} \quad (3.2)$$

where

$$m_\epsilon(\lambda) = -\ln \beta(\lambda) - \ln(\mu_d + \epsilon) > 0.$$

Lemma 3.2 (A priori bound). *There exists a constant $C_1 = C_1(s, \rho) < \infty$ such that*

$$\mathbb{E}_x(|G_{\omega,\lambda}(x, x; z)|^s) \leq C_1 \lambda^{-s} \quad (3.3)$$

for all $x \in \mathbb{Z}^d$, $z \in \mathbb{C} \setminus \mathbb{R}$, and $\lambda > 0$.

Here

$$C_1 = \frac{1}{1-s}$$

for $\omega_x \sim \mathcal{U}[-1, 1]$

Lemma 3.3. RESOLVENT IDENTITY *cómo en el libro*

Proof of a priori bound. This proof demonstrates the simplicity of the fundamental idea underlying the FMM. For fixed $x \in \mathbb{Z}^d$, write $\omega = (\hat{\omega}, \omega_x)$ where $\hat{\omega}$ is short for $(\omega_u)_{u \in \mathbb{Z}^d \setminus \{x\}}$. With $P_{e_x} := \langle e_x, \cdot \rangle e_x$, the orthogonal projection onto the span of e_x , we can separate the ω_x and $\hat{\omega}$ dependence of H_ω

as

$$H_\omega = -\Delta + \lambda V_{\hat{\omega}} + \lambda \omega_x P_{e_x} = H_{\hat{\omega}} + \lambda \omega_x P_{e_x}.$$

For the operators $(H_\omega - z)$ and $(H_{\hat{\omega}} - z)$, the resolvent identity yields

$$(H_\omega - z)^{-1} = (H_{\hat{\omega}} - z)^{-1} - \lambda \omega_x (H_{\hat{\omega}} - z)^{-1} P_{e_x} (H_\omega - z)^{-1}. \quad (3.4)$$

Taking matrix-elements we conclude for the corresponding diagonal Green functions that

$$G_{\omega,\lambda}(x, x; z) = G_{\hat{\omega},\lambda}(x, x; z) - \lambda \omega_x G_{\hat{\omega},\lambda}(x, x; z) G_{\omega,\lambda}(x, x; z) \quad (3.5)$$

and

$$G_{\omega,\lambda}(x, x; z) = \frac{1}{a + \lambda \omega_x} \quad \text{with } a = \frac{1}{G_{\hat{\omega},\lambda}(x, x; z)}. \quad (3.6)$$

Note that the latter is well-defined since one can easily check that $\text{Im } G_{\hat{\omega},\lambda}(x, x; z) > 0$ for $\text{Im } z > 0$.

The important fact is that a is a complex number which does not depend on ω_x . Thus, writing $\mathbb{E}_x(\dots) := \int \dots \rho(\omega_x) d\omega_x$ for $\rho(\omega_x) = \frac{1}{2} \chi_{[-1,1]}$, we find that

$$\mathbb{E}_x(|G_{\omega,\lambda}(x, x; z)|^s) = \frac{1}{2\lambda^s} \int_{-1}^1 \frac{d\omega_x}{|\frac{a}{\lambda} + \omega_x|^s} \leq \frac{1}{2\lambda^s} \int_{-1}^1 \frac{d\omega_x}{|\omega_x|^s} = \frac{1}{(1-s)\lambda^s}, \quad (3.7)$$

with $\frac{1}{1-s}$ independent of λ and a , and thus independent of $\hat{\omega}$, z and x . $\square$

3.2.1 From fractional moment's exponential bound to dynamical localisation

In this subsection we aim to prove the following theorem:

Theorem 3.4. *Let $I \subset \mathbb{R}$ be an open bounded interval. If there exist $s \in (0, 1)$, $C < \infty$ and $\mu > 0$ such that*

$$\mathbb{E}(|G(x, y; E + i\epsilon)|^s) \leq C e^{-\mu|x-y|} \quad (3.8)$$

uniformly in $E \in I$ and $\epsilon > 0$, then dynamical localization holds on the interval I .

Before diving in the proof, we are going to need more equipment.

Lemma 3.5. *For a bounded and self-adjoint operator A ,*

$$\|(A - z)^{-1}\| \leq \frac{1}{\text{Im}(z)} \quad (3.9)$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$.

Proof. For the definition of the operator norm, we know that

$$\|(A - z)^{-1}\| := \sup_{\mu \in \sigma((A - z)^{-1})} |\mu|$$

Let's consider a $\lambda \in \mathbb{C}$ and write $\mu = \frac{1}{\lambda - z}$. This means:

$$\begin{aligned} \frac{1}{\lambda - z} \in \sigma((A - z)^{-1}) &\iff (A - z)^{-1} - (\lambda - z)^{-1} \text{ not invertible} \\ &\iff (A - z)^{-1}(A - \lambda)(\lambda - z)^{-1} \text{ not invertible} \\ &\iff (A - \lambda) \text{ not invertible} \\ &\iff \lambda \in \sigma(A) \end{aligned}$$

Where the resolvent identity has been used in third Implication.

At this point we obtain

$$\|(A - z)^{-1}\| = \sup_{\lambda \in \sigma(A)} \frac{1}{|\lambda - z|} = \frac{1}{\inf_{\lambda \in \sigma(A)} |\lambda - z|} = \frac{1}{\text{dist}(\sigma(A), z)}$$

For a bounded and self-adjoint operator, we know that the spectrum is real-valued. For $z \in \mathbb{C} \setminus \mathbb{R}$ the minimum distance between z and $\sigma(A)$ is the imaginary component of z :

$$\|(A - z)^{-1}\| \leq \frac{1}{|\text{Im}(z)|} \quad (3.10)$$

□

This bound is one of the most important and useful in this thesis. Once again we applied the resolvent identity to find a simple yet useful result.

Proposition 3.6. *For every $s \in (0, 1)$ there exists a constant $C_1 < \infty$ only depending on s and ρ such that*

$$|\text{Im } z| \mathbb{E}_x (|G(x, y; z)|^2) \leq C_1 \mathbb{E}_x (|G(x, y; z)|^s) \quad (3.11)$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$ and $x, y \in \mathbb{Z}^d$.

Proof. As in the proof of Lemma 3.2 we write $\omega = (\hat{\omega}, \omega_x)$. Keep $\hat{\omega}$ fixed and consider the Hamiltonian

$$H^{(\alpha)} = H(\hat{\omega}, \omega_x + \alpha) = H_\omega + \alpha P_{e_x}.$$

Its Green function will be denoted by $G^{(\alpha)}$. Applying the resolvent identity, we find

$$(H_\omega - z)^{-1} = (H^{(\alpha)} - z)^{-1} + \alpha(H_\omega - z)^{-1}P_{e_x}(H^{(\alpha)} - z)^{-1}$$

and

$$\begin{aligned} G^{(\alpha)}(x, y; z) &= \frac{G_\omega(x, y; z)}{1 + \alpha G_\omega(x, x; z)} \\ &= \frac{1}{\alpha + G_\omega(x, x; z)^{-1}} \cdot \frac{G_\omega(x, y; z)}{G_\omega(x, x; z)}. \end{aligned} \quad (3.12)$$

For the special case $x = y$ and $\tilde{\alpha} = -\operatorname{Re} G_\omega(x, x; z)^{-1}$ we get from (3.11) and Lemma 3.5 that

$$\left| \frac{1}{\operatorname{Im} G(x, x; z)^{-1}} \right| = |G^{(\tilde{\alpha})}(x, x; z)| \leq \frac{1}{|\operatorname{Im} z|},$$

i.e. $|\operatorname{Im} G(x, x; z)^{-1}| \geq |\operatorname{Im} z|$. Inserting this into (3.11) gives

$$|\operatorname{Im} z| |G^{(\alpha)}(x, y; z)|^2 \leq \frac{|\operatorname{Im} G_\omega(x, x; z)^{-1}|}{|\alpha + G_\omega(x, x; z)^{-1}|^2} \cdot \frac{|G_\omega(x, y; z)|^2}{|G_\omega(x, x; z)|^2}. \quad (3.13)$$

On the other hand, we can bound the same expression by

$$\begin{aligned} |\operatorname{Im} z| |G^{(\alpha)}(x, y; z)|^2 &\leq |\operatorname{Im} z| \sum_{y' \in \mathbb{Z}^d} |G^{(\alpha)}(x, y'; z)|^2 \\ &= |\operatorname{Im} z| \langle e_x (H^{(\alpha)} - z)^{-1}, (H^{(\alpha)} - z)^{-1} e_x \rangle \\ &= |\operatorname{Im} z| \langle e_x, (H^{(\alpha)} - \bar{z})^{-1} (H^{(\alpha)} - z)^{-1} e_x \rangle \\ &= |\operatorname{Im} z| \langle e_x, \frac{1}{z - \bar{z}} \left[(H^{(\alpha)} - z)^{-1} - (H^{(\alpha)} - \bar{z})^{-1} \right] e_x \rangle \\ &= |\operatorname{Im} G^{(\alpha)}(x, x; z)| \\ &= \frac{|\operatorname{Im} G_\omega(x, x; z)^{-1}|}{|\alpha + G_\omega(x, x; z)^{-1}|^2}, \end{aligned} \quad (3.14)$$

where the third step used the symmetry of $H^{(\alpha)}$ and last step used (3.11) with $x = y$.

For $t \geq 0$ one has $\min(1, t^2) \leq t^s$, with $s \in (0, 1)$. Using this to interpolate between (3.12) and (3.13) we get

$$|\operatorname{Im} z| |G^{(\alpha)}(x, y; z)|^2 \leq \frac{|\operatorname{Im} G_\omega(x, x; z)^{-1}|}{|\alpha + G_\omega(x, x; z)^{-1}|^2} \cdot \frac{|G_\omega(x, y; z)|^s}{|G_\omega(x, x; z)|^s}. \quad (3.15)$$

We will now use the following “re-sampling trick”, which has the effect of creating an additional random variable (here α) to average over. For a non-negative Borel function f on $\mathbb{R}$,

$$\begin{aligned} \iint f(\omega_x + \alpha) \rho(\omega_x + \alpha) d\alpha \rho(\omega_x) d\omega_x &= \iint f(\omega_x + \alpha) \rho(\omega_x + \alpha) \rho(\omega_x) d\omega_x d\alpha \\ &= \iint f(\omega_x) \rho(\omega_x) \rho(\omega_x - \alpha) d\omega_x d\alpha \\ &= \int f(\omega_x) \rho(\omega_x) \left(\int \rho(\omega_x - \alpha) d\alpha \right) d\omega_x \\ &= \int f(\omega_x) \rho(\omega_x) d\omega_x, \end{aligned} \quad (3.16)$$

where in the first and third steps we used Fubini and in the second we used translation invariance of Lebesgue measure.

Choose $f(\omega_x) = |G_{(\hat{\omega}, \omega_x)}(x, y; z)|^2$; then (3.14) and (3.15) yield

$$\begin{aligned} |\operatorname{Im} z| \mathbb{E}_x(|G_\omega(x, y; z)|^2) &= |\operatorname{Im} z| \mathbb{E}_x \left(\int |G^{(\alpha)}(x, y; z)|^2 \rho(\omega_x + \alpha) d\alpha \right) \\ &\leq \mathbb{E}_x \left(|\operatorname{Im} G_\omega(x, x; z)^{-1}| \frac{|G_\omega(x, y; z)|^s}{|G_\omega(x, x; z)|^s} \int \frac{\rho(\omega_x + \alpha)}{|\alpha + G_\omega(x, x; z)^{-1}|^2} d\alpha \right). \end{aligned} \quad (3.17)$$

We now use Lemma 3.7 below with $w = G_\omega(x, x; z)^{-1}$ to conclude

$$|\operatorname{Im} z| \mathbb{E}_x(|G_\omega(x, y; z)|^2) \leq C \mathbb{E}_x(|G_\omega(x, y; z)|^s),$$

with a constant $C < \infty$ which only depends on $\operatorname{supp} \rho$, but not on x, y and z . $\square$

Lemma 3.7. *Let $0 < s < 1$. There exists a constant $C'_s < \infty$ such that*

$$|\operatorname{Im} w| \cdot |w|^s \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha \leq C'_s$$

uniformly in $w \in \mathbb{C}$ and $\omega_x \sim \mathcal{U}[-1, 1]$.

Proof. The proof is that of [REFERENCE GUNTER STOLZ]. Since $|w|^s \leq |\alpha|^s + |\alpha + w|^s$, we bound:

1.

$$\begin{aligned} |\operatorname{Im} w| \int \frac{|\alpha|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha &\leq |\operatorname{Im} w| \int \frac{\| |\alpha|^s \rho(\omega_x + \alpha) \|_\infty}{|\alpha + w|^2} d\alpha \\ &\leq |\operatorname{Im} w| \| |\alpha|^s \rho(\omega_x + \alpha) \|_\infty \int \frac{1}{(\alpha + \operatorname{Re} w)^2 + \operatorname{Im} w^2} d\alpha \end{aligned}$$

for $a, b \in \mathbb{R}$ we have

$$\int \frac{1}{a^2 + b^2} da = 2\pi i \operatorname{Res} \left(\frac{1}{a^2 + b^2}, ib \right) = \frac{2\pi i}{2ib}.$$

Thus for $\rho = \frac{1}{2}\chi_{[-1,1]}$ and $|\alpha| \leq 1 + |\omega| \leq 2$

$$\begin{aligned} |\operatorname{Im} w| \int \frac{|\alpha|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha &\leq \pi \| |\alpha|^s \rho(\omega_x + \alpha) \|_\infty \\ &\leq \pi 2^{s-1}. \end{aligned}$$

2. Now, we bound the second expression.

(a) For $|\operatorname{Im} w|$ big:

Since $|\alpha + w| \geq |\operatorname{Im} w|$ and $\int \rho(\omega_x + \alpha) d\alpha = 1$:

$$I_2 \leq |\operatorname{Im} w| \cdot \frac{1}{|\operatorname{Im} w|^{2-s}} \int \rho(\omega_x + \alpha) d\alpha = \frac{1}{|\operatorname{Im} w|^{1-s}}$$

(b) For $|\operatorname{Im} w|$ small:

$$\begin{aligned} |\operatorname{Im} w| \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^{2-s}} d\alpha &\leq |\operatorname{Im} w| \frac{1}{2} \int \frac{d\alpha}{|\alpha + w|^{2-s}} \\ &= |\operatorname{Im} w| \frac{1}{2} \int \frac{1}{|u + \operatorname{Im} w|^{2-s}} du \\ &= |\operatorname{Im} w|^2 \frac{1}{2} \int \frac{1}{|u^2 + \operatorname{Im} w^2|^{\frac{2-s}{2}}} du \\ &= \frac{1}{2} (\operatorname{Im} w)^s \int \frac{1}{|t^2 + 1|^{\frac{2-s}{2}}} dt \end{aligned}$$

by setting $u = \alpha + \operatorname{Re} w$ and $t = \frac{u}{\operatorname{Im} w}$.

To establish the finiteness of the constant, we consider the integral

$$C_s := \int_{-\infty}^{\infty} \frac{1}{(t^2 + 1)^{\frac{2-s}{2}}} dt.$$

For $s \in (0, 1)$, the exponent satisfies $2 - s > 1$, which ensures the asymptotic decay $(t^2 + 1)^{-\frac{2-s}{2}} \sim |t|^{-(2-s)}$ as $|t| \rightarrow \infty$, rendering the integral absolutely convergent. Then we have

$$|\operatorname{Im} w| \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^{2-s}} d\alpha \leq \min \left(\frac{1}{|\operatorname{Im} w|^{1-s}}, C_s \frac{1}{2} |\operatorname{Im} w|^s \right) \leq C'_s 2^{s-1}.$$

□

Now we can proceed with the proof of the theorem.

Proof of Theorem 3.4. Let $\mu_{x,y}$ be the complex Borel spectral measure of h_ω associated with vectors δ_x, δ_y , defined by

$$\mu_{x,y}(B) := \langle \delta_x, \chi_B(h_\omega) \delta_y \rangle.$$

The total variation measure $|\mu_{x,y}|$ over the open bounded interval I satisfies

$$\sup_{t \in \mathbb{R}} |\langle \delta_x, e^{-it h_\omega} \chi_I(h_\omega) \delta_y \rangle| \leq |\mu_{x,y}|(I).$$

By Lusin's Theorem, $|\mu_{x,y}|(I) = \sup_{g \in C_c(I), \|g\|_\infty \leq 1} |\langle \delta_x, g(h_\omega) \delta_y \rangle|$. For every $g \in C_c(I)$, the spectral calculus and Parseval's identity yield

$$\langle \delta_x, g(h_\omega) \delta_y \rangle = \lim_{\epsilon \downarrow 0} \frac{\epsilon}{\pi} \int_I g(E) \langle \delta_x, (h_\omega - E - i\epsilon)^{-1} (h_\omega - E + i\epsilon)^{-1} \delta_y \rangle dE.$$

Inserting a partition of unity I :

$$|\langle \delta_x, g(h_\omega) \delta_y \rangle| \leq \liminf_{\epsilon \downarrow 0} \frac{\epsilon}{\pi} \int_I \sum_{z \in \mathbb{Z}^d} |G_\omega(x, z; E + i\epsilon)| |G_\omega(z, y; E + i\epsilon)| dE.$$

Taking expectations, applying Fatou's Lemma, Fubini's Theorem, and the

Cauchy–Schwarz inequality on the expectation gives:

$$\begin{aligned}\mathbb{E}(|\mu_{x,y}|(I)) &\leq \frac{1}{\pi} \liminf_{\epsilon \downarrow 0} \int_I \sum_{z \in \mathbb{Z}^d} \mathbb{E}(\epsilon |G_\omega(x, z; E + i\epsilon)| |G_\omega(z, y; E + i\epsilon)|) dE \\ &\leq \frac{1}{\pi} \liminf_{\epsilon \downarrow 0} \int_I \sum_{z \in \mathbb{Z}^d} (\mathbb{E}(\epsilon |G_\omega(x, z; E + i\epsilon)|^2))^{1/2} (\mathbb{E}(\epsilon |G_\omega(z, y; E + i\epsilon)|^2))^{1/2} dE.\end{aligned}$$

Applying Proposition to both factors and then substituting the uniform hypothesis:

$$\begin{aligned}\mathbb{E}(|\mu_{x,y}|(I)) &\leq \frac{C_1}{\pi} \int_I \sum_{z \in \mathbb{Z}^d} (\mathbb{E}|G_\omega(x, z; E + i\epsilon)|^s)^{1/2} (\mathbb{E}|G_\omega(z, y; E - i\epsilon)|^s)^{1/2} dE \\ &\leq \frac{C_1 |I| C}{\pi} \sum_{z \in \mathbb{Z}^d} e^{-\frac{\mu}{2}|x-z|} e^{-\frac{\mu}{2}|z-y|}.\end{aligned}$$

By the triangle inequality, $\frac{\mu}{2}|x-z| + \frac{\mu}{2}|z-y| \geq \frac{\mu}{4}|x-y| + \frac{\mu}{4}|x-z| + \frac{\mu}{4}|z-y|$. Therefore,

$$\mathbb{E}(|\mu_{x,y}|(I)) \leq \left(\frac{C_1 |I| C}{\pi} \sum_{z \in \mathbb{Z}^d} e^{-\frac{\mu}{4}|x-z|} e^{-\frac{\mu}{4}|z-y|} \right) e^{-\frac{\mu}{4}|x-y|} \leq \tilde{C} e^{-\tilde{\mu}|x-y|},$$

with $\tilde{\mu} = \frac{\mu}{4}$ and finite prefactor $\tilde{C} = \frac{C_1 |I| C}{\pi} \sum_{u \in \mathbb{Z}^d} e^{-\frac{\mu}{4}|u|}$. $\square$

Remark 3.8 (Spectral Localization). Via the RAGE theorem, spectral localisation is also implied.

4 SAW representation

4.1 SAW Definitions

Definition 4.1 (Self-Avoiding Walk). A **self-avoiding walk** (SAW) on the integer lattice $\mathbb{Z}^d$ of length $N \in \mathbb{N}_0$ is an ordered $(N + 1)$ -tuple of distinct lattice sites

$$\gamma = (\gamma_0, \gamma_1, \dots, \gamma_N) \in (\mathbb{Z}^d)^{N+1}$$

such that consecutive vertices are nearest neighbors, that is,

$$|\gamma_i - \gamma_{i-1}| = 1 \quad \text{for all } i \in \{1, \dots, N\},$$

and $\gamma_i \neq \gamma_j$ for all $i \neq j$.

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Definition 4.2 (Sets of Self-Avoiding Walks). Let $\Lambda \subseteq \mathbb{Z}^d$. For any $x, y \in \Lambda$ and $N \in \mathbb{N}_0$:

- (i) $\mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N contained entirely in Λ , starting at $\gamma_0 = x$ and terminating at $\gamma_N = y$. When $\Lambda = \mathbb{Z}^d$, we simply write $\mathcal{S}_N(y, x)$.
- (ii) $\mathcal{S}_N^{(\Lambda)}(x) := \bigcup_{y \in \Lambda} \mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N in Λ starting at x .
- (iii) $c_N := |\mathcal{S}_N(x)| = \sum_{y \in \mathbb{Z}^d} |\mathcal{S}_N(y, x)|$ denotes the total number of self-avoiding walks of length N on $\mathbb{Z}^d$ originating from a fixed site (e.g., the origin 0), which is independent of the starting vertex by translation invariance.

Definition 4.3 (Connectivity Constant). The **connective constant** (or connectivity constant) μ_d of the lattice $\mathbb{Z}^d$ is defined as the asymptotic growth rate of c_N :

$$\mu_d := \lim_{N \rightarrow \infty} (c_N)^{1/N}.$$

By subadditivity of $\log c_N$, this limit exists, is positive, and satisfies the strict inequality $\mu_d < 2d - 1$.

Claim 4.4. Let μ_d be the connective constant (or growth rate) of self-avoiding walks on $\mathbb{Z}^d$. Then:

$$\mu_d \leq 2d - 1$$

Proof. Let c_N denote the number of self-avoiding walks (SAWs) of length N starting from the origin $0 \in \mathbb{Z}^d$, and let $S_N(0, x)$ denote the set of such walks ending at $x \in \mathbb{Z}^d$. By definition:

$$c_N = \sum_{x \in \mathbb{Z}^d} |S_N(0, x)| = \sum_{x \in \mathbb{Z}^d} \sum_{\gamma \in S_N(0, x)} 1$$

Consider the generating function (*susceptibility*) with parameter $\xi \in \mathbb{C}$:

$$\sum_{N=0}^{\infty} |\xi|^N c_N = \sum_{N=0}^{\infty} |\xi|^N \sum_{x \in \mathbb{Z}^d} |S_N(0, x)|$$

For $N = 0$, $c_0 = 1$. For $N \geq 1$, a walk cannot immediately step back onto the previous vertex (non-reversing condition). Thus:

- The first step has at most $2d$ choices.
- Each of the subsequent $N - 1$ steps has at most $2d - 1$ choices.

This gives the upper bound:

$$c_N \leq 2d(2d - 1)^{N-1}, \quad \text{for } N \geq 1.$$

Using this upper bound, we can bound the series from above:

$$\begin{aligned} \sum_{N=0}^{\infty} |\xi|^N c_N &\leq 1 + \sum_{N=1}^{\infty} |\xi|^N (2d)(2d - 1)^{N-1} \\ &= 1 + 2d|\xi| \sum_{k=0}^{\infty} |\xi|^k (2d - 1)^k \quad (\text{setting } k = N - 1) \\ &= 1 + 2d|\xi| \sum_{k=0}^{\infty} (|\xi|(2d - 1))^k \end{aligned}$$

This geometric series converges if and only if:

$$|\xi|(2d - 1) < 1 \iff |\xi| < \frac{1}{2d - 1} =: R$$

where $R = \frac{1}{2d-1}$ is the radius of convergence of the bounding series.

By the Cauchy-Hadamard theorem, the radius of convergence R' of the original series $\sum_{N=0}^{\infty} c_N |\xi|^N$ is:

$$R' = \frac{1}{\limsup_{N \rightarrow \infty} c_N^{1/N}} = \frac{1}{\mu_d}$$

Since the original series is bounded term-by-term by a series that converges for $|\xi| < R$, the true radius of convergence must satisfy:

$$R' \geq R \implies \frac{1}{\mu_d} \geq \frac{1}{2d-1}$$

Taking the reciprocal yields:

$$\mu_d \leq 2d - 1.$$

□

Definition 4.5 (SAW Correlation Function). For a parameter $\beta \geq 0$, the **self-avoiding walk correlation function** $C_\beta(y - x)$ between two vertices $x, y \in \mathbb{Z}^d$ is the generating function

$$C_\beta(y - x) := \sum_{N=0}^{\infty} \beta^N |\mathcal{S}_N(y, x)|,$$

defined for all values of β for which the power series is absolutely convergent. By translation invariance, $C_\beta(y - x)$ depends only on the displacement vector $y - x$.

Definition 4.6 (SAW Susceptibility). The **susceptibility** $\chi(\beta)$ of self-avoiding walks is the sum of the two-point correlation function over all end-points:

$$\chi(\beta) := \sum_{x \in \mathbb{Z}^d} C_\beta(x) = \sum_{N=0}^{\infty} c_N \beta^N.$$

The series **converges uniformly absolutely** if and only if $0 \leq |\beta|1/\mu_d$, where $1/\mu_d$ is the critical radius of convergence.

Proposition 4.7 (Exponential Decay of the Correlation Function). *Let $\beta \geq 0$ be such that $\beta < 1/\mu_d$. Then for every $\varepsilon > 0$ there exists a constant $K_\varepsilon < \infty$, independent of x , such that*

$$C_\beta(x) \leq K_\varepsilon ((\mu_d + \varepsilon)\beta)^{|x|} \quad \text{for all } x \in \mathbb{Z}^d.$$

Proof. We know that $|\mathcal{S}_N(0, x)| = 0$ whenever $N < |x|$, and the sum defining $C_\beta(x)$ may be restricted to $N \geq |x|$:

$$C_\beta(x) = \sum_{N=|x|}^{\infty} \beta^N |\mathcal{S}_N(x, 0)| \leq \sum_{N=|x|}^{\infty} \beta^N c_N. \quad (4.1)$$

Fix $\varepsilon > 0$. Since $\mu_d = \lim_{N \rightarrow \infty} c_N^{1/N}$, by the definition of limit there exists $N_0 \in \mathbb{N}$ such that

$$c_N^{1/N} < \mu_d + \varepsilon \quad \text{for all } N \geq N_0,$$

that is,

$$c_N < (\mu_d + \varepsilon)^N \quad \text{for all } N \geq N_0.$$

For $N = 0, 1, \dots, N_0 - 1$, define

$$K_\varepsilon := \max \left(1, \max_{0 \leq N < N_0} \frac{c_N}{(\mu_d + \varepsilon)^N} \right),$$

which is finite, being the maximum of finitely many finite quantities. By construction it holds:

$$c_N \leq K_\varepsilon (\mu_d + \varepsilon)^N \quad \text{for all } N \geq 0.$$

Combining it with (4.1) we obtain

$$C_\beta(x) \leq K_\varepsilon \sum_{N=|x|}^{\infty} [(\mu_d + \varepsilon)\beta]^N.$$

Choosing ε small enough that $r := (\mu_d + \varepsilon)\beta < 1$ (possible since $\beta < 1/\mu_d$), the tail of the geometric series equals

$$\sum_{N=|x|}^{\infty} r^N = \frac{r^{|x|}}{1 - r}.$$

Hence

$$C_\beta(x) \leq \frac{K_\varepsilon}{1 - (\mu_d + \varepsilon)\beta} [(\mu_d + \varepsilon)\beta]^{|x|}.$$

where $\frac{K_\varepsilon}{1 - (\mu_d + \varepsilon)\beta}$ is still finite and x -independent. □

4.2 SAW representation of $\mathbf{G}$

The proof of the theorem 3.1 is similar to the one given by Schenker in [1]. We are going to work with the Green's function defined on a volume where one site is removed every stage. Therefore, for any $\Lambda \supset \mathbb{Z}^d$ (finite or infinite)

the Green function changes:

$$G^\Lambda(x, y; z) := \begin{cases} (H_\omega^\Lambda - z), & \forall x, y \in \Lambda \\ 0 & , \text{otherwise} \end{cases} \quad (4.2)$$

where $H_\omega^\Lambda - z \in \mathbb{C}^{\Lambda \times \Lambda}$ is the matrix obtained by restricting $H_\omega - z$ to Λ . In this thesis we will consider $\Lambda = \mathbb{Z}^d$, but we leave the notation Λ to highlight the fact that the same result holds for any volume.

To notice is also that $H_\omega^\Lambda - z$ is still invertible for all $z \in \mathbb{C} \setminus \mathbb{R}$. In this section, for simplicity during calculations, we will write $G^\Lambda(x, y)$ instead of $G^\Lambda(x, y; z)$.

Theorem 4.8. *Let $x, y \in \mathbb{Z}^d$, $x \neq y$, $\Lambda \subseteq \mathbb{Z}^d$. G^Λ is defined as in (4.1). It holds*

$$G^\Lambda(x, y) = \sum_{n=1}^N \sum_{\gamma \in \mathcal{S}_n(x, y)} \prod_{i=1}^n G^\Lambda(x, x) \cdot G^{\Lambda_i}(\gamma_i, \gamma_i) + R_N \quad (4.3)$$

where $\Lambda_i = \Lambda \setminus \{x, \gamma_1, \dots, \gamma_{i-1}\}$ and

$$R_N = \sum_{x \sim y} \sum_{\gamma \in \mathcal{S}_N(0, x)} \prod_{i=1}^{N-1} G^\Lambda(x, x) \cdot G^{\Lambda_i}(\gamma_i, \gamma_i) \cdot G^{\Lambda_N}(\gamma_N, y). \quad (4.4)$$

Claim 4.9.

$$G^\Lambda(x, y) = -G^\Lambda(x, x) \sum_{x_1 \sim x} G^{\Lambda_1}(x_1, y) \quad (4.5)$$

Proof. Using the resolvent identity for $A = (H_\omega^\Lambda - z)$ and $B = (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)$, we obtain:

$$\begin{aligned} & (H_\omega^\Lambda - z)^{-1} - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \\ &= -(H_\omega^\Lambda - z)^{-1} \left(H_\omega^\Lambda - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}}) \right) (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \end{aligned}$$

We know:

$$\begin{aligned} H_\omega^\Lambda &= 2d - \sum_{x_1 \sim y} \delta_{x_1} + \lambda V_\omega \quad \forall y \in \Lambda \\ H_\omega^{\{x\}} &= \lambda \omega_x + 2d \\ H_\omega^{\Lambda \setminus \{x\}} &= 2d - \sum_{x_1 \sim y} \delta_{x_1} + \lambda V_\omega - \lambda \omega_x \quad \forall y \in \Lambda \setminus \{x\} \end{aligned}$$

$$\implies H_\omega^\Lambda - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}}) = - \sum_{x_1 \sim x} \delta_{x_1} =: T_{x, x_1}$$

Now considering the matrix elements in (x, y) with $x \neq y$ and matrix multiplication:

$$(H_\omega^\Lambda - z)_{x, y}^{-1} - 0 = - \sum_{x_1, x_2} (H_\omega^\Lambda - z)_{x, x_1}^{-1} T_{x_1, x_2} (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)_{x_2, y}^{-1}$$

For $T_{x_1, x_2} \neq 0$, we need $x_1 = x$ and $x_2 \sim x_1$.

We obtain:

$$G^\Lambda(x, y) = \sum_{x_1 \sim x} G^\Lambda(x, x) G^{\Lambda \setminus \{x\}}(x_1, y)$$

Now we separate the case in which $x_1 = y$ (which implies $|x - y| = 1$) from the cas in which it isn't.

$$G^\Lambda(x, y) = G^\Lambda(x, x) \left(\sum_{x_1 \sim x, x_1 \neq y} G^{\Lambda_1}(x_1, y) + I_{|x-y|=1} G^{\Lambda_1}(y, y) \right)$$

Iterating N times, it gives (4.2) □

4.3 Proof with SAW

Now we use self-avoiding-walks to prove the theorem (3.1).

Proof. We take the average of (4.2), and using $|\sum a|^s \leq \sum |a|^s$ (which follows from triangle inequality), we have:

$$\mathbb{E} [|G^\Lambda(x, y)|^s] \leq \sum_{n=1}^N \sum_{\gamma \in \mathcal{S}_n(x, y)} \mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{i=1}^n |G^{\Lambda_i}(\gamma_i, \gamma_i)|^s \right] + \quad (4.6)$$

$$\sum_{x \sim y} \sum_{\gamma \in \mathcal{S}_N(0, x)} \mathbb{E} [|G^\Lambda(x, x)|^s \prod_{i=1}^{N-1} |G^{\Lambda_i}(\gamma_i, \gamma_i)|^s \cdot |G^{\Lambda_N}(\gamma_N, y)|^s] \quad (4.7)$$

The resolvent $G^{\Lambda_j}(w, w')$ does not depend on the random variables ω_{x_i} , $i = 0, \dots, j - 1$. Hence, recursive applications of the a priori bound (3.2) yield

$$\mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{j=1}^n |G^{\Lambda_j}(\gamma_j, \gamma_j)|^s \right] \leq \left(\frac{1}{1-s} \frac{1}{\lambda^s} \right)^{n+1}$$

and

$$\begin{aligned} & \mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{j=1}^{N-1} |G^{\Lambda_j}(\gamma_j, \gamma_j)|^s \cdot |G^{\Lambda_{N-1}}(\gamma_N, y)|^s \right] \\ & \leq \left(\frac{1}{1-s} \cdot \frac{1}{\lambda^s} \right)^N \cdot \mathbb{E} [|G^{\Lambda_{N-1}}(\gamma_N, y)|^s] \leq \left(\frac{1}{1-s} \frac{1}{\lambda^s} \right)^N \frac{1}{|\operatorname{Im}(z)|^s} \end{aligned}$$

Where we used a priori bound and lemma 3.5.

Inserting these estimates in the sums above, we get:

$$\mathbb{E} \left(|G^{(\Lambda)}(x, y)|^s \right) \leq \sum_{n=0}^N \beta(s)^{1+n} |\mathcal{S}_n^{(\Lambda)}(y, x)| + \beta(s)^N |\mathcal{S}_N^{(\Lambda)}(x)| \frac{1}{|\operatorname{Im} z|^s} \quad (4.8)$$

with $\beta(s) = \frac{1}{1-s} \frac{1}{\lambda^s}$. Let's consider the susceptibility $\chi(\beta)$ of the self avoiding walks in $\mathcal{S}_N^{(\Lambda)}(x)$:

$$\chi(\beta) = \sum_{x \in \mathbb{Z}^d} \sum_{N=0}^{\infty} \beta(s)^N |\mathcal{S}_N^{(\Lambda)}(x)| = \sum_{N=0}^{\infty} c_N \beta(s)^N. \quad (4.9)$$

The series converges absolutely if and only if $0 \leq |\beta| < 1/\mu_d$, where $1/\mu_d = \lim_{N \rightarrow \infty} c_N^{1/N}$ is the critical radius of convergence. Thus, for $|\beta(s)| = \frac{1}{|(1-s)\lambda^s|} \leq \frac{1}{\mu_d}$, $\chi(\beta)$ converges uniformly, and most importantly:

$$c_N \beta(s)^N \longrightarrow 0, \quad \text{for } N \rightarrow \infty$$

We send N to the infinity and obtain from (4.7):

$$\mathbb{E} \left(|G^{(\Lambda)}(x, y)|^s \right) \leq \sum_{n=0}^{\infty} \beta(s)^{1+n} |\mathcal{S}_n^{(\Lambda)}(y, x)| \quad (4.10)$$

Taking the supremum over volumes Λ now yields

$$\sup_{\Lambda \subset \mathbb{Z}^d} \mathbb{E} \left(|G^{(\Lambda)}(x, y)|^s \right) \leq \beta(s) C_{\beta(s)}(x - y). \quad (4.3)$$

It remains to establish for which λ we may find $s \in (0, 1)$ with $\beta(s) < 1/\mu_d$. As above, for $\lambda > e$ the unique critical point is $s_{\text{crit}} = 1 - 1/\ln \lambda$ and

$$\min_{s \in (0, 1)} \beta(s) = \beta(s_{\text{crit}}) = \frac{e \ln \lambda}{\lambda}$$

which is less than $1/\mu_d$ if and only if $\lambda > \lambda_{\text{And}}$. Finally we get:

$$\mathbb{E} \left(|G^{(\Lambda)}(x, y; z)|^{1 - \frac{1}{\ln \lambda}} \right) \leq \beta(\lambda) K_\epsilon \ln \lambda e^{-m_\epsilon(\lambda)|x-y|} \quad (4.11)$$

where

$$m_\epsilon(\lambda) = -\ln \beta(\lambda) - \ln(\mu_d + \epsilon) > 0.$$

□